ETHICAL HACKING

PRACTICAL HANDBOOK

FINAL EDITION • 16 TOOLS • KALI LINUX VM + WINDOWS HOST

Commands with explanations, placeholder discovery, Windows/Kali alternatives, and click-by-click GUI workflows.

Important: Use only your own isolated lab or instructor-authorized targets. This PDF is a practical guide, not proof that

commands have been executed on your exact computer. Verify options with each tool's local help.

Before you start: find the values inside example values

Replace the entire placeholder, including < and >. Use the example values below only as a guide; your actual lab values may differ.

Kali network discovery

```
cat /etc/os-release
ip -br addr
ip route
ip link
ip neigh
```

Example only: if Kali is 192.168.56.101/24 and the lab target is 192.168.56.102, use eth0 and 192.168.56.0/24 only if your own output confirms them.

Find target IP

• Linux/Metasploitable console: ip -br addr or ip addr; ifconfig may also be available.

• Windows lab target: ipconfig.

• Kali and target should share the intended Host-only/Internal Network.

After you identify the real target:

```
ping -c 4 192.168.56.102
ip route get 192.168.56.102
```

Windows host checks (PowerShell)

```
Get-NetIPConfiguration
Get-NetAdapter
Get-NetRoute -AddressFamily IPv4
ipconfig /all
route print
arp -a
```

Placeholder lookup

Placeholder How to find it

eth0 Kali: ip -br addr and ip route; choose interface connected to lab.

192.168.56.0/24 Kali: ip route; use connected lab network prefix.

192.168.56.102 Target console: ip addr/ifconfig or Windows ipconfig.

example.com Example used in commands: example.com. Use only domains assigned by your instructor for real testing.

http://192.168.56.102 Use exact URL of local/approved app from lab instructions.

/home/kali/captures/lab.pcapng Kali: pwd; ls -lh; find ~/ -type f -iname '*.pcap*' 2>/dev/null. Windows: File Explorer.

DC=lab,DC=local/dc.lab.local Get from instructor/directory admin; don't guess.

/etc/snort/snort.conf//etc/snort/rules/local.rules Inspect package output and /etc/snort; use actual installed path.

>

credentials/community Use instructor-issued lab values only; don't guess or expose secrets.

1. Kali Linux VM and lab setup

Run in: Kali terminal + VirtualBox/VMware + target console

Prepare isolated lab and verify network connectivity.

Commands

Kali

```
cat /etc/os-release
ip -br addr
ip route
ip neigh
ping -c 4 192.168.56.102
ip route get 192.168.56.102
```

Only on authorized lab network:

```
sudo nmap -sn 192.168.56.0/24
```

Windows option / checks

```
ipconfig /all
Get-NetIPConfiguration
route print
arp -a
```

GUI / workflow

• Power off both VMs → Settings → Network → choose Host-only or Internal Network; use the same lab network for both VMs.

• Start both. Get Kali IP with ip -br addr and target IP from target console.

• If no connectivity, check adapter mode, subnet, DHCP and target power state.

Important: Do not bridge a vulnerable VM to a public/production network. Example subnet 192.168.56.0/24 is not universal.

2. Sysmon

Run in: Windows PowerShell as Administrator + Event Viewer

Install Sysmon from Microsoft Sysinternals and review process-creation events.

Commands

In folder containing Sysmon64.exe and approved XML:

.\Sysmon64.exe -?

.\Sysmon64.exe -accepteula -i .\sysmonconfig.xml

.\Sysmon64.exe -c

```
Get-Service Sysmon64,Sysmon -ErrorAction SilentlyContinue
Get-WinEvent -FilterHashtable @{
```

LogName='Microsoft-Windows-Sysmon/Operational'; Id=1

} -MaxEvents 20 | Select TimeCreated,Id,Message

wevtutil epl Microsoft-Windows-Sysmon/Operational .\Sysmon-Operational.evtx

GUI / workflow

• Extract Sysmon, open PowerShell as Administrator in that folder, and use instructor-approved sysmonconfig.xml.

• Win+R → eventvwr.msc → Applications and Services Logs → Microsoft → Windows → Sysmon → Operational.

• Filter Current Log for ID 1 (process creation). ID 3 (network connection) requires configuration.

Important: The XML and executable must exist in the current directory. If the event channel is missing, check install output.

3. theHarvester

Run in: Kali terminal

Collect publicly indexed emails/hosts/subdomains for the assigned domain; sources vary by version.

Commands

```
sudo apt update
sudo apt install theharvester -y
theHarvester --version
theHarvester -h
theHarvester -d example.com -b crtsh
theHarvester -d example.com -b bing
theHarvester -d example.com -f harvester-results
```

Important: Choose a source shown by your local -h output. Sources may require API keys, be rate-limited, or change; zero

results proves nothing.

4. Netdiscover

Run in: Kali terminal; local Layer-2 lab segment

Find live devices with ARP on your authorized lab subnet.

Commands

```
sudo apt update
sudo apt install netdiscover -y
netdiscover -h
ip -br addr
ip route
sudo netdiscover -i eth0 -r 192.168.56.0/24
sudo netdiscover -p -i eth0
sudo netdiscover -i eth0 -r 192.168.56.0/24 2>&1 | tee netdiscover.txt
```

Important: -i selects interface; -r selects CIDR range; -p passive mode on common versions. Confirm options with netdiscover -h.

5. Google Dorking

Run in: Browser search box; not terminal

Search publicly indexed results within the assigned domain. Do not access private data.

Commands

```
site:example.com
site:example.com filetype:pdf
site:example.com filetype:docx
site:example.com filetype:xlsx
site:example.com intitle:"index of"
site:example.com inurl:login
site:example.com inurl:backup
site:example.com "confidential"
```

GUI / workflow

• Replace example.com with assigned scope and paste one query at a time into Google.

• Record query, result URL, date and apparent exposure. Do not attempt logins or exploit findings.

6. FOCA

Run in: Windows FOCA GUI

Find public documents and inspect metadata such as author, software, timestamps and local paths.

Windows option / checks

```
Get-Date
```

Test-NetConnection www.google.com -Port 443

Test-NetConnection www.bing.com -Port 443

```
Resolve-DnsName www.google.com
```

GUI / workflow

• Extract FOCA and read its release requirements; old builds may have dependency/search-parser problems.

• Launch FOCA → Project → New Project (labels vary) → enter project name and assigned domain.

• Choose available search engines/file types → search → select permitted public documents → run metadata analysis.

• Review author/creator, application, timestamps, usernames and local paths; save project/report/screenshots.

• For 429/parser errors, record limitation and retry later or use another available provider; don't claim success without results.

Important: FOCA is a GUI tool, not a standard Kali CLI command. SQL Server compatibility depends on the exact FOCA build.

7. WHOIS

Run in: Kali terminal; Windows browser/client

Look up public domain/IP registration details.

Commands

```
sudo apt update
sudo apt install whois -y
whois --version
whois example.com
whois example.com | tee whois-output.txt
dig example.com NS
dig example.com SOA
```

Windows option / checks

```
Resolve-DnsName example.com -Type NS
Resolve-DnsName example.com -Type MX
```

For WHOIS/RDAP, use reputable registry/registrar lookup in browser.

Important: Registrant data may be redacted. A redacted field is not proof of maliciousness.

8. nslookup

Run in: Windows CMD/PowerShell and Kali

Query DNS records and compare resolvers.

Commands

```
nslookup example.com
nslookup -type=A example.com
nslookup -type=AAAA example.com
nslookup -type=MX example.com
nslookup -type=NS example.com
nslookup example.com 1.1.1.1
```

Kali alternative

```
dig +short example.com A
dig +short example.com MX
dig +short example.com CNAME
```

Windows option / checks

```
nslookup microsoft.com
nslookup -type=MX microsoft.com
nslookup -type=NS microsoft.com
Resolve-DnsName microsoft.com -Type A
```

Important: Use assigned domain for exam work; microsoft.com here is only a public syntax example.

9. Sublist3r

Run in: Kali terminal

Enumerate public subdomains for the assigned domain; older tool and sources may be incompatible.

Commands

```
sudo apt update
sudo apt install sublist3r -y
sublist3r -h
python3 --version
command -v sublist3r
sublist3r -d example.com
sublist3r -d example.com -o subdomains.txt
cat subdomains.txt
```

Important: If package/source fails, document the error and follow instructor-approved alternative. Do not switch target scope.

10. Maltego

Run in: Maltego desktop GUI

Build a graph linking domains, DNS names, IPs, emails and organizations.

GUI / workflow

• Launch Maltego and sign in if required; transforms depend on account/plan/data sources.

• Create graph → add Domain entity for assigned domain → select entity → inspect available transforms.

• Run permitted transforms, inspect sources, save/export graph and capture screenshots.

Important: Graph links are leads, not proof of common ownership; independently verify important relationships.

11. Shodan

Run in: Browser

Review indexed observations of public assets; results can be stale and do not authorize access.

Commands

Paste into Shodan website search:

```
hostname:example.com
domain:example.com
hostname:example.com port:443
hostname:example.com port:80
ssl.cert.subject.cn:example.com
```

GUI / workflow

• Search only assigned domain/IP scope.

• Record reported ports, product/banner, last-seen context and date; save screenshots if permitted.

• Do not log into, probe or exploit a service discovered in results.

Important: Filters/export/API access depend on account. Shodan is a browser service, not a Kali command.

12. Nmap

Run in: Kali; Windows if installed

Discover hosts and inspect ports/services on authorized lab targets.

Commands

```
sudo apt update
sudo apt install nmap -y
nmap --version
nmap --help
nmap -sn 192.168.56.0/24
nmap 192.168.56.102
nmap -p 22,80,443 192.168.56.102
nmap -sV 192.168.56.102
nmap -oN nmap-report.txt 192.168.56.102
nmap -oA nmap-report 192.168.56.102
```

Windows option / checks

```
nmap --version
nmap -sn 192.168.56.0/24
nmap -sV 192.168.56.102
```

Test-NetConnection 192.168.56.102 -Port 80

Important: -sn = host discovery; -p = port selection; -sV = service/version detection; -oN = text output; -oA = multiple formats.

See official Nmap reference: https://nmap.org/book/man.html

13. Nessus

Run in: Nessus web GUI; Windows host if installed there

Run vulnerability assessment and export report for assigned lab target.

Windows option / checks

```
Get-Service *nessus* -ErrorAction SilentlyContinue
```

Test-NetConnection localhost -Port 8834

GUI / workflow

• Install Nessus from Tenable's official distribution and complete activation/initialization.

• Open local dashboard (commonly https://localhost:8834).

• Create suitable scan template → enter only assigned target → configure credentials only if approved → launch.

• Review severity, evidence and remediation; export report with scan date and target.

Important: If dashboard won't open, check service status and port 8834. URL alone doesn't mean Nessus is installed.

14. Wireshark

Run in: Wireshark GUI on Windows/Kali; tshark in Kali

Analyze supplied packet capture or permitted lab traffic.

Commands

```
wireshark --version
tshark --version
tshark -D
tshark -r /home/kali/captures/lab.pcapng
tshark -r /home/kali/captures/lab.pcapng -Y dns
tshark -r /home/kali/captures/lab.pcapng -Y 'tcp.port == 443'
tshark -r /home/kali/captures/lab.pcapng -T fields -e frame.time -e ip.src -e ip.dst -e _ws.col.Protocol
```

Windows option / checks

Open capture: Wireshark → File → Open

Display filters:

dns

http

icmp

ip.addr == 192.168.56.102

tcp.port == 443

GUI / workflow

• Open supplied .pcap/.pcapng via File → Open; for live capture choose lab interface → Start → Stop.

• Apply display filter; inspect packet details; use Follow → TCP Stream where relevant; save evidence.

• Windows live capture may require Npcap.

Important: Find the actual capture path using pwd/ls or File Explorer. Do not capture other people's traffic.

15. Robtex

Run in: Browser; Kali for cross-checks

Review public DNS/IP relationships and historical data available on Robtex.

Commands

```
dig example.com A
dig example.com NS
dig example.com MX
dig -x 192.168.56.102
whois example.com
```

GUI / workflow

• Open Robtex in browser and search assigned domain or IP.

• Review available DNS, related IP/domain and historical data; record source/date.

• Cross-check against authoritative DNS/WHOIS and capture screenshots.

Important: Robtex is a website, not a standard terminal command. Shared hosting/history does not prove common ownership.

16. Armitage / Metasploit

Run in: Kali; isolated vulnerable VM only

Armitage is legacy and may not run on current Kali. Use lab snapshots and instructor-approved modules only.

Commands

```
msfconsole --version
msfconsole -h
command -v armitage
sudo systemctl status postgresql
```

GUI / workflow

• Take VM snapshot and verify vulnerable target is isolated.

• Follow installed release documentation to start services/GUI; exact launch method varies.

• Connect framework, add only assigned target, review services, and use only instructor-approved lab module.

• Record result and restore snapshot after exercise.

Important: Do not assume a systemd service named metasploit exists; names differ by release. If Armitage is unavailable, ask if

console Metasploit is permitted.

Exam-day checklist

• Identify the correct environment for each command: Kali, Windows PowerShell/CMD, or GUI/browser.

• Run ip -br addr and ip route; identify the lab interface/subnet and target IP.

• Replace all example values with real authorized values and remove angle brackets.

• Check local -h/--help for version-sensitive options.

• Save command output, screenshots, reports and dates.

• Only test assigned targets; restore lab snapshots and undo temporary changes.

Quick troubleshooting

• Command not found: check installation/PATH; Kali: command -v nmap; PowerShell: Get-Command nmap.

• Package unavailable: sudo apt update; apt-cache policy netdiscover; check Kali repository/network.

• No hosts found: check VM adapter mode, interface, subnet, target power and firewall.

• Empty OSINT result: provider limits, rate limits or API keys may explain it; record limitation.

• GUI unavailable: check installation, service status and documented local port.